

MossWater: A Living Media Interface for Encouraging Office Workers' Daily Water Intake

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ABSTRACT

Tracking drinking behaviors and digitally presenting the information can motivate people's daily water intake. In this paper, we present MossWater, a living interface designed to visualize the daily water intake and so as to motivate people to drink water more frequently and proactively. The system utilizes living interface as an unobtrusive ambient display to evoke empathy. Caring and watering the moss is designed as a metaphor for caring for people's health by regularly drinking, which invokes people's feeling of empathy to motivate their healthier behaviors. This paper also presents the results of a user study exploring the usability of MossWater and users' feedbacks to this living interface. Our system is expected to promote the welfare of both human and nature in the long-term.

CCS CONCEPTS

• Human-centered computing; • Human computer interaction (HCI); • Ubiquitous and mobile computing;

KEYWORDS

Living media interface, water intake, ambient display, empathy, plant-based displays

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1 INTRODUCTION

Water is an essential nutrient and has numerous roles in the human body, such as the building material of cells and the carrier of nutrients in organs and tissues [10]. Mild dehydration can induce significant impairments in both cognitive function and physical

performance [18, 19]. Therefore, regular and sufficient water intake is essential for people to keep healthy. However, many people tend to forget to drink water when busy with work. As notifications may cause distractions and interruptions for workers [17], there is a need to design a system to remind people to drink water in their working environments but in a non-intrusive way.

Plants are ubiquitous in environments and have an irreplaceable role in our daily life, such as purifying the air and relieving visual fatigue. In the field of Human-Computer Interactions, researchers have utilized plants to create novel interfaces. Ambient Intelligence [1] refers to invisible, embedded technologies in our natural surroundings with effortless and straightforward interactions. By integrating living plants, researchers could create natural and unobtrusive applications [7, 14, 15]. As humans only have true empathy for living creatures [20], researchers tried to use plants to create living media interfaces to invoke empathy and induce behavior change [5, 6].

In this paper, we present MossWater, a living media interface using a metaphor to encourage office workers' daily water intake (Figure 1). Watering moss is triggered by people's drinking behaviors. Dry moss reflects water loss of the human body. As moss's spontaneous dehydration is perceivable [12], we use this phenomenon to make people be aware of their water demands and motivate them to drink water. Caring for the plant on the desk has been proved to improve office workers' mental health and maintain their interest over time [21]. Our system is a healing interface using people's acts of caring for plants to achieve long-term attraction. Through this system, users care for themselves by caring for the moss.

The main contributions of our work are: 1) involving empathy in motivating people to drink water. 2) designing a living interface with spontaneous changes of moss. 3) visualizing people's daily water intake and remind them in a non-intrusive way.

2 RELATED WORKS

Our work builds on previous researches towards HCI designs for motivating water intake and plant-based ambient displays.

2.1 HCI designs for motivating water intake

In recent years, HCI researchers have shown interest in solving the problem of insufficient water intake in people's daily life. Mug-tree [13] is a concept consisting of a smart mug and a digital photo frame displaying a water drinking game. It inspired empathy by using users' acts of caring for a virtual tree as a metaphor. Similarly,

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Figure 1: MossWater

WaterCoaster [16] also uses a metaphor to present feedbacks to drinking behaviors and evoke empathy. In its mobile game application, the water level of a fish tank and the facial expressions of an underwater creature are both related to users' water intake. However, they both give feedbacks through traditional screen-based displays that may distract attentions from users' work. Moreover, they just use virtual images, cultivating less empathy compared with living creatures.

Ambient displays are increasingly used to present water intake information to give abstract, non-intrusive and aesthetic feedbacks. GROW [11] is a smart bottle utilizing its surface as an ambient display to encourage daily water intake. It presents a tree image made of thermo-chromic print on the bottle surface when the user's water intake is detected by a liquid level sensor. Another example is Wwall [22], a smart water dispenser system giving collective feedback and individual feedback through projected displays. It records the data of individual water intake and presents it via a portion of the full image to motivate individuals to fulfill the collective goals of water intake. Our work also uses an ambient display to visualize the drinking behaviors of users without disturbing them.

2.2 Plant-based ambient displays

Plants are constantly accumulating and displaying the effect of exterior stimuli over time [8]. As the stimuli such as light [8], water [9] and nutrition [4] can be manipulated by diverse actuators, consequent changes in the appearance of plants are utilized by HCI researchers as kinds of displays of specific information. Infotropism [8] uses lights to manipulate the growth direction of corn seedlings to display the difference in contributions on recycling or littering of people. In addition to manipulating the growth direction, the growth rate of plants is also utilized as another display mechanism.

The water for Meet Eater [9] is decided by the level of social interaction on the website, while the nutrient concentration of Biogotchi [4] depends on the collection of users' step data each day.

As incorporating living media interface can create empathy [3], plant-based interfaces show the potential of motivating positive change in people's behaviors. Go and grow [2] is a living visualization system with cyclamen. The water amount of the cyclamen is linked to the progress of the user's daily step, and Biogotchi [4] has a similar concept. Both of them show the emotional connections that traditional digital devices do not have. Another example is Raffigh [6], a living media interface designed to motivate children to play learning games. It encourages children care for a living mushroom as the duration of water is based on the amount of children's input speeches. In summary, plant-based interfaces can create empathy and display information in a manipulative and non-intrusive way. In this work, we extend the plant-based ambient display for motivating daily water intake.

3 SYSTEM DESIGN

Different from the previous researches, we aim to use plants as a living interface that tracks information and arouses users' empathic feelings to motivate their daily water intake. To achieve this, which plant to be used and how the information is presented are two major design consideration issues. We choose a kind of moss called *Hypnum plumaeforme* as the living interface. It closes leaves in a dehydration state and opens in a sufficient water state, providing a difference in appearance. The moss will be watered and become moist when the user drinks water, and it will constantly dry when not watered. The state of moss is used to reminds users to drink water as an ambient display. In addition, the water sound (natural and relaxing) will be played as another ambient reminder (perceivable and unobtrusive) if users neglect the change of moss.

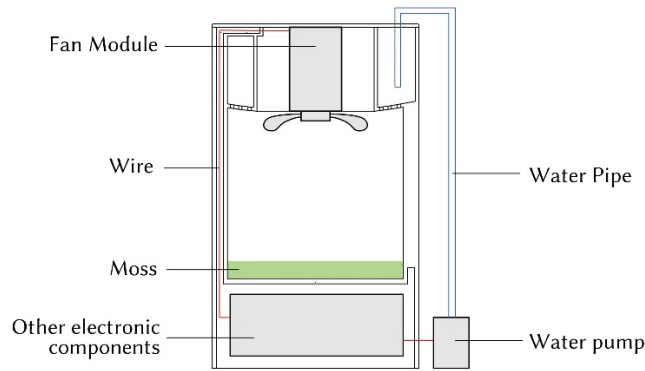


Figure 2: Outline of the moss-based display

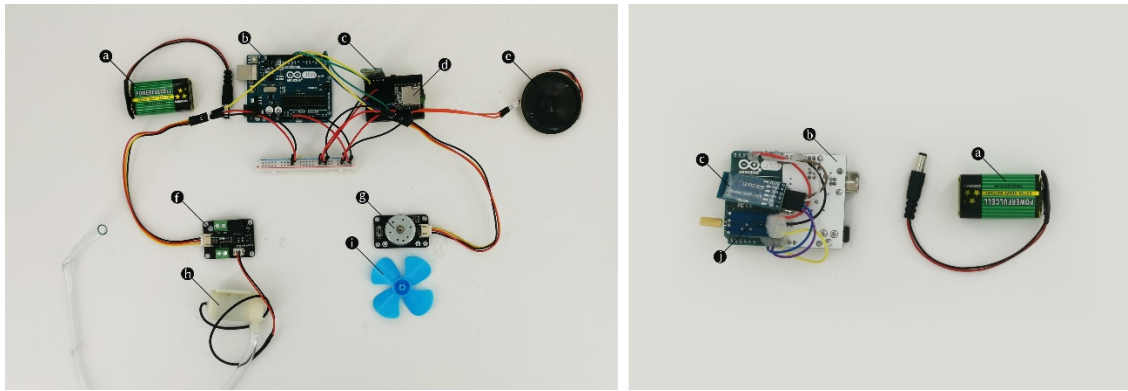


Figure 3: Electronic components of MossWater. (a) Battery (b) Arduino microcontroller (c) HC-05 Bluetooth Module (d) DF-Player Mini (e) Loudspeaker (f) YwRobot Motor drive module T0 (g) EZ Fan Module R300 (h) Water pump (i) Fan (j) SW-520D tilt sensor

We designed three main functions of the MossWater system: 1) drying the moss within a certain period to remind users to drink water; 2) watering the moss when users' drink behaviors are detected; 3) using a natural and soft sound as another auditory reminder if users don't react for an extended period.

3.1 The components of the system

We developed MossWater with quick prototyping technologies such as Arduino and 3D-printing. The system consists of two parts. One is a moss-based display that waters itself and plays sounds based on the users' drinking behaviors (Figure 2). The other is a removable device on the cup that detects users' behaviors of drinking water and sends the results to the former.

In the first part, moss is planted in a 3D-printed pot. Except for the living element, it consists of an Arduino microcontroller, three different actuators (a water pump, a fan, and a voice module), a Bluetooth module and a power cell. As the display is designed to be unobtrusive and fit on the desk, all electronic components are

placed in the interior of the container with only the water pump in the external water source. The water pump is connected with a water pipe and a wire, which are separately connected to an annular water channel on the top of the container and the drive module on the bottom of the container. The fan is stuck at the center of the cover. It's used to air dry the moss so that the time of moss dehydration can be modified to remind users to drink water at their expected time. The voice module is composed of an MP3 player and a loudspeaker, while the sound played is stored locally. This module is placed at the bottom of the container as well as the Arduino microcontroller, the Bluetooth module and the power cell. The Bluetooth in there is the master device in the master-slave communication mode.

In the second part, the device is attached to a magic tape so that it can be removed to any cup. It consists of an Arduino microcontroller, a tilt sensor, a Bluetooth module and a power cell. All of these components are wrapped in a case. The Bluetooth in there is the slave device that sends signals to the master device through Arduino Software Serial based on detection the tilt sensor's detection results.



Figure 4: Four stages of MossWater. (a) MossWater is watering when the user drinks water (b) MossWater in a sufficient water state (c) MossWater is air-dried and turns into dehydration (d) MossWater plays a soft water sound

3.2 The usage of the system

The pot is placed on the user's desk, while the detection device is tied to the user's cup. The fan on the top of the pot is kept open to dry the moss within 60 minutes, which is the expected time interval between users' drinking behaviors. The time is calculated based on three parameters: a sedentary adult should drink 1.5L of water per day [10], the average 8 hours work time (China's legal working hours), assumed amount of 100ml water intake at a single time.

When the user drinks water, the tilt sensor on the cup will be triggered and send a signal through the Bluetooth slave device. Once the Bluetooth master device in the pot receives the signal, the water pump will be opened for 4 seconds to water the moss while the fan will be interrupted for 5 minutes to avoid splashing or disturbing the process of moss absorbing water (Fig. 4(a)). As the moss absorbs the water and open leaves, it turns into a sufficient water state and presents greener (Fig. 4(b)). Then, the moss will be air-dried constantly with the help of the fan until it turns into dehydration (Fig. 4(c)). As mentioned previously, the process takes 60 minutes. The dehydration state (yellowed) is used to remind users to drink water. However, if the user drinks water within an interval, the moss will be watered again as a new starting point. Otherwise, a soft water sound will be played by the voice module and gradually turned up to a perceivable volume as another reminder (Fig. 4(d)). We also set a 5 minutes interval for the water pump, which means if the user drinks water very frequently in a short time (less likely to be dehydrated), overwatering of the moss also needs to be prevented as protection of moss.

4 PRELIMINARY USER STUDY

4.1 Participants and Study Procedure

To get people's reactions to this concept and examine the usability of the prototype, we carried out a preliminary user study with five office workers (three female and two male) who have sedentary works in Internet companies of Hangzhou. The user study was carried out separately in their workplaces. Their ages varied between 24 to 27 (*Mean* = 25.2, *SD* = 1.36).

Semi-structured interviews and observations were conducted during the process of the user test. At the beginning, participants were given the explanation of MossWater. Then, we asked them to place the functional prototype on their desks with the detection device attaching to their cups. Participants were also asked to use the system only during their working hours. For the interviews, we asked them some open-ended questions to find the effects of the living interface and their usage experiences.

4.2 Findings from the user study

Based on the observation and interview responses, we found that the participants' reactions to MossWater were positive. On the one hand, most participants expressed that using this device could obtain a novel experience of being reminded to drink water while not disturbing their work. They couldn't help associating moss with drinking water, which gave them a sense of responsibility. E.g., "When I inadvertently see the dehydrated state of a moss, I realize that I need to drink some water" and "the sound reminds me that the moss is waiting for me to take care of it". On the other hand, they indicated that this device provided an enjoyable experience by the interactable plants. Four of them expressed that

they would temporarily divert their attention to watch the watering process making them feel relaxed while plants were being watered. Moreover, they expected the consequent watering of the device after drinking water. Besides, there was an interesting finding that two of the participants naturally touched the plants without being told. Most participants expressed that they were willing to put such a device on their desktop as it was good-looking and functional.

However, they also indicated several concerns and suggestions. One participant was concerned about the repeated watering and drying may cause a negative effect on the moss. Another participant mentioned that he couldn't form an emotional connection with the plant during the working hours of one day. Regarding the design of prototypes, they suggested adding a function that could adjust the interval of reminding according to their own situation. Some participants wished to get some rewards for themselves while drinking water in addition to watering moss. They also suggested that using more moss for display would be better.

4.3 Discussion

Most participants in our study shares a positive attitude to our concept and willingness for long-term use. Moreover, the results of the study give evidences that living plant has the potential to become an ambient and empathic interface encouraging people to drink water in working environments. We also believe that the living media interface could bring unexpected experiences (such as touching nature), which is also supported by the study.

On the other hand, based on the comments of potential users, the system needs to be further improved to support better performance. To be more specific, the water intake at a time does vary from different individuals. As the time interval between drinking needs to be calculated based on that, we intend to obtain it by detecting the intake every time or letting users set the amount in advance. The system's feedback could also be enriched to provide better user experiences, such as involving olfactory feedback to please users. As we only conducted the user test with each participant during one working day, following up user study needs to be carried out to prove their positive roles and attractions in the long-term.

5 CONCLUSION AND FUTURE WORK

We presented MossWater, an empathic living media interface capable of detecting personal drinking behaviors and aims to motivate individuals' daily water intake through an ambient display based on plants in working environments. In our system, we chose moss because it could provide a difference in its appearance based on water. We presented the design and implementation of the system. Moreover, we confirmed MossWater could be a potential system of inspiring users' emotional responses (such as empathy, pleasure) to motivate drinking behaviors through the results of a preliminary user study.

In the future, we plan to revise our prototype and combine it with new feedback mechanisms based on the results of the preliminary user study. We will conduct a follow-up study with a sufficient number of participants in the long-term to investigate the effectiveness with the new iteration and explore the framework of involving living plants as a unique interface in designs for the HCI field.

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